

CHAGANTI TRAYI

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Full-Stack & ML Engineer | Java, Spring Boot, Angular, AWS | AI/ML | REST/GraphQL | Jenkins CI/CD

SUMMARY

Full-Stack ML Engineer with 3+ years of experience building and modernizing enterprise applications using **Spring Boot, Angular, and AWS**. Specialized in **AI/ML model development, data engineering, and intelligent automation**, integrating predictive analytics into production-grade applications using **Python, TensorFlow, Scikit-learn, and AWS SageMaker**. Develops **Microservices**, event-driven systems with **Kafka**, and cloud-native engineering using **Docker, Kubernetes, and Terraform**. Skilled in REST/GraphQL APIs, real-time dashboards, SQL programming, and database optimization. Adept in CI/CD automation, Agile/Scrum practices, and delivering secure, scalable, high-performance solutions.

SKILLS

- **Programming Languages:** Java, SQL, TypeScript, PL/SQL
 - **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Keras, NLTK, SpaCy
 - **Cloud & MLOps:** AWS, Docker, Kubernetes, MLflow, TensorBoard
 - **Web Technologies:** HTML5, CSS3, JavaScript, Angular, CodeIgniter, RESTful APIs
 - **Cloud & DevOps:** AWS (EC2, S3, VPC, RDS, DynamoDB, Lambda), Docker, Kubernetes, Jenkins
 - **Databases:** SQL Server, PostgreSQL, MySQL
 - **Messaging/Streaming:** Apache Kafka, RabbitMQ
 - **Security:** OAuth2, JWT, Spring Security
 - **Frameworks & Libraries:** Spring Boot, Spring Data JPA, Spring MVC, Spring Cloud, Hibernate, React.js, Node.js
 - **Tools & Platforms:** IntelliJ IDEA, Visual Studio, Android Studio, GitHub Actions, Maven, Git, Gradle, JIRA, Postman, Swagger
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EXPERIENCE

CVS Health - Software Development Engineer | Sep 2024 – May 2025 | Texas, USA

- Architected 10+ Spring Boot and Node.js microservices for the insurance claims lifecycle, enabling secure and scalable REST/GraphQL APIs.
- Designed event-driven workflows with Apache Kafka, processing 250K+ daily claim events with 99.9% uptime.
- Implemented OAuth2.0 with JWT tokens and enforced mTLS, eliminating unauthorized API access incidents.
- Developed Angular 16 dashboards with NgRx state management, reducing page load times by 70% and improving customer experience.
- Implemented small-scale AI/ML models for prediction, classification, or recommendation.
- Built real-time analytics dashboards with D3.js visualizations, enabling leadership to track KPIs and claims lifecycle trends.
- Modernized legacy AngularJS modules to Vue.js, reducing technical debt and improving maintainability by 30%.
- Developed ML-powered anomaly detection to flag fraudulent insurance claims, leveraging Scikit-learn Random Forest models with 92% precision.
- Communicated ML insights and pipeline performance metrics to technical stakeholders and non-technical business teams.
- Built and maintained Java-ML-based systems for data analysis and predictive modeling, ensuring smooth integration with existing Java codebases and frameworks for media content analysis.
- Implemented end-to-end AI/ML solutions for model development, validation, and deployment within Java environments using H2O.ai, enhancing scalability and performance for media analytics applications.
- Integrated trained ML models into production APIs via Flask and AWS Lambda, reducing false-positive claims by 20%.
- Built a predictive analytics dashboard in Angular using D3.js and Chart.js for real-time visualization of model predictions.
- Designed and developed JavaFX-based user interfaces to facilitate real-time monitoring and analysis of AI models, improving usability and content management for media professionals.

- Deployed and managed machine learning models using AI Platform and TensorFlow on Google Cloud, ensuring robust and scalable AI solutions.
- Automated CI/CD deployment pipelines for ML APIs using Jenkins and Terraform, ensuring continuous delivery on AWS ECS.
- Collaborated with data engineers to create ETL pipelines that prepared 10M+ records daily for ML training using AWS Glue and Pandas.
- Containerized ML microservices using Docker and orchestrated workloads through Kubernetes for scalable inference.

CognitiveBotics Technologies - Associate Software Developer | Jan 2023 – June 2023 | Hyderabad, India

- Developed Spring Boot microservices for product features, independently deployed on AWS EC2, reducing downtime by 25%.
- Built REST APIs with Swagger/OpenAPI documentation, streamlining integration for frontend and mobile teams.
- Designed reporting dashboards using Spring MVC + Hibernate, improving reporting accuracy by 40%.
- Implemented Angular 12 component-driven UIs with RxJS streams, reducing app crashes by 20%.
- Integrated ML recommendation module using Flask API for personalized product suggestions based on user interactions.
- Developed scalable AI/ML applications using Java, integrating Spring for robust backend services, and Hibernate for efficient ORM and database management, enhancing data handling and performance.
- Built REST APIs with Swagger/OpenAPI documentation, streamlining integration for frontend and mobile teams.
- Automated CI/CD pipelines with Jenkins and GitHub Actions, reducing deployment cycles from 2 days to 4 hours.
- Configured Dockerized ML and API services, enabling faster and consistent deployment across environments.
- Collaborated with data scientists to operationalize trained ML models within microservices architecture.
- Engineered RESTful APIs with Spring Boot, enabling seamless integration of AI/ML models into applications and ensuring reliable data exchange and interaction across diverse systems.
- Architected microservices using Spring Cloud, facilitating modular AI/ML solutions that support high scalability and efficient deployment of machine learning models.
- Developed a Sales Forecasting Engine using XGBoost and LSTM, increasing forecast accuracy by 35% and integrated results into a web-based dashboard using Flask + Angular.
- Implemented computer vision pipeline for defect detection using OpenCV and TensorFlow, reducing manual inspection time by 40%.
- Automated model retraining workflows using AWS SageMaker Pipelines and MLflow for experiment tracking and model versioning.
- Deployed real-time inference APIs on AWS Lambda and monitored latency and throughput using CloudWatch metrics.
- Collaborated with multidisciplinary teams to embed AI insights into enterprise dashboards and analytics tools.

Honeywell – Jr Software Developer | June 2020 – Dec 2022 | Hyderabad, India

- Engineered backend Java services consuming REST APIs on AWS EC2, improving data exchange efficiency by 45%.
- Modelled domain entities with Hibernate associations and mappings, reducing DB query times by 30%.
- Led migration from AngularJS to Angular 8, improving frontend performance and maintainability.
- Built Vue.js SPAs with Router/Vuex, enhancing user experience and enabling modular UI development.
- Implemented CI/CD pipelines with Jenkins, Maven, Ansible, Terraform, reducing production deployment errors by 50%.
- Implemented data access and transaction management in AI-driven applications using Hibernate, streamlining data persistence and retrieval for optimized machine learning workflows.
- Developed an AI-driven fault prediction system using Random Forests on historical telemetry data to predict equipment failure with 89% accuracy.
- Implemented CI/CD pipelines with Jenkins, Maven, and Ansible, reducing manual deployment efforts by 50%.
- Integrated machine learning insights into dashboards, helping maintenance teams make proactive operational decisions.

EDUCATION

Master of Science in Computer Science

University of Central Missouri, Lee's Summit| GPA: 3.71

Bachelor of Science in Computer Science Engineering

Blekinge Institute of Technology, Karlskrona, Sweden| GPA: 3.4

Bachelor of Technology in Computer Science Engineering

Jawaharlal Nehru Technological University Kakinada (JNTUK), Kakinada, India| GPA: 3.4